

3. A chromosome mutation on the X chromosome can result in the blood clotting disorder haemophilia A.

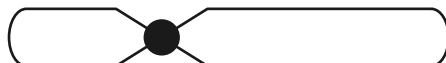
The diagram shows two chromosomes from an individual with haemophilia A.

X chromosome



mutation area

Y chromosome



- (a) (i) Name the type of inheritance for haemophilia A.

1

- (ii) In a population, 6300 males have haemophilia A. This represents 75% of the total number of individuals with this condition.

Calculate how many females have haemophilia A in this population.

1

Space for calculation

[Turn over



3. (continued)

- (b) The diagram shows the order of exons in the mutation area of the X chromosome from an individual with haemophilia A and an individual without haemophilia A.

with haemophilia A

22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	23	24	25	26
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1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
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without haemophilia A

- (i) Name the type of chromosome mutation that causes haemophilia A. 1
- _____
- (ii) State the function of an exon. 1
- _____
- (iii) During the expression of a gene for a blood clotting protein, mRNA is transcribed from exons 1 to 26. 2
- Explain why this protein is non-functional in individuals with haemophilia A.
- _____
- _____
- _____
- (c) Describe how alternative RNA splicing leads to different proteins being expressed from one gene. 1
- _____
- _____

